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Beyond Common Factors?

The Predictive Value of Sparse Information in Euro-Area Forecasting

Appendix

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A Supplementary Exhibits

Note. The paper can be read without this appendix, which is supplied as a separate file. It collects supporting exhibits cited in the text (the data catalogue, factor-model diagnostics, details of the forecasting ladder and robustness checks); it adds no argument of its own. Unless a note says otherwise, the forecasts are scored on target months to December 2025 without 2020, and the source is EA-MD-QD and the external series of Table A1, own calculations.

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A.1 Data

Table A1: Series in the panel, by information set

Code	Description	Set	Block	TR3	Delay	Source
UNETOT_EA	Unemployment: Total (% active population)	M	Labor market	5	2	Eurostat
UNEO25_EA	Unemployment: Over 25 years (% active population)	M	Labor market	5	2	Eurostat
UNEU25_EA	Unemployment: Under 25 years (% active population)	M	Labor market	5	2	Eurostat
REER42_EA	Real Exchange Rate (42 main industrial countries)	M	Financial markets	5	2	Eurostat
ERUS_EA	Exchange Rate (US dollar)	M	Financial markets	5	1	ECB
SHIX_EA	Stock Price Index	M	Financial markets	2	1	OECD
IRT3M_EACC	3-Months Interest Rates	M	Interest rates	4	1	Eurostat
IRT6M_EACC	6-Months Interest Rates	M	Interest rates	4	1	Eurostat
LTIRT_EACC	Long-Term Interest Rates (EMU Criterion)	M	Interest rates	4	1	Eurostat
IPMN_EA	Industrial Production Index: Manufacturing	M	IP and turnover	2	2	Eurostat
IPCAG_EA	Industrial Production Index: Capital Goods	M	IP and turnover	2	2	Eurostat
IPCOG_EA	Industrial Production Index: Consumer Goods	M	IP and turnover	2	2	Eurostat
IPDCOG_EA	Industrial Production Index: Durable Consumer Goods	M	IP and turnover	2	2	Eurostat
IPNDCOG_EA	Industrial Production Index: Non Durable Consumer Goods	M	IP and turnover	2	2	Eurostat
IPING_EA	Industrial Production Index: Intermediate Goods	M	IP and turnover	2	2	Eurostat
IPNRG_EA	Industrial Production Index: Energy	M	IP and turnover	2	2	Eurostat
TRNMN_EA	Turnover Index: Manufacturing	M	IP and turnover	2	3	Eurostat
TRNCAG_EA	Turnover Index: Capital Goods	M	IP and turnover	2	3	Eurostat
TRNCOG_EA	Turnover Index: Consumer Goods	M	IP and turnover	2	3	Eurostat
TRNDCOG_EA	Turnover Index: Durable Consumer Goods	M	IP and turnover	2	3	Eurostat

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Table A1: Series in the panel, by information set

Code	Description	Set	Block	TR3	Delay	Source
TRNNDCOG_EA	Turnover Index: Non Durable Consumer Goods	M	IP and turnover	2	3	Eurostat
TRNING_EA	Turnover Index: Intermediate Goods	M	IP and turnover	2	3	Eurostat
TRNNRG_EA	Turnover Index: Energy	M	IP and turnover	2	3	Eurostat
CAREG_EA	Passenger's Cars Registrations	M	IP and turnover	2	1	ECB
PPICAG_EA	Producer Price Index: Capital Goods	M	Prices	2	2	Eurostat
PPICOG_EA	Producer Price Index: Consumer Goods	M	Prices	2	2	Eurostat
PPINDCOG_EA	Producer Price Index: Non Durable Consumer Goods	M	Prices	2	2	Eurostat
PPIDCOG_EA	Producer Price Index: Durable Consumer Goods	M	Prices	2	2	Eurostat
PPIING_EA	Producer Price Index: Intermediate Goods	M	Prices	2	2	Eurostat
PPINRG_EA	Producer Price Index: Energy	M	Prices	2	2	Eurostat
HICPOV_EA	Harmonized Index of Consumer Prices: Overall Index	M	Prices	2	1	ECB
HICPNEF_EA	Harmonized Index of Consumer Prices: All Items, no Energy&Food	M	Prices	2	1	ECB
HICPG_EA	Harmonized Index of Consumer Prices: Goods	M	Prices	2	1	ECB
HICPIN_EA	Harmonized Index of Consumer Prices: Industrial Goods	M	Prices	2	1	ECB
HICPSV_EA	Harmonized Index of Consumer Prices: Services	M	Prices	2	1	ECB
HICPNG_EA	Harmonized Index of Consumer Prices: Energy	M	Prices	2	1	Eurostat
ICONFIX_EA	Industrial Confidence Indicator	M	Confidence	4	0	Eurostat
CCONFIX_EA	Consumer Confidence Index	M	Confidence	4	0	Eurostat
ESENTIX_EA	Economic Sentiment Indicator	M	Confidence	4	0	Eurostat
KCONFIX_EA	Construction Sentiment Indicator	M	Confidence	4	0	Eurostat
RTCONFIX_EA	Retail Confidence Indicator	M	Confidence	4	0	Eurostat
SCONFIX_EA	Services Confidence Indicator	M	Confidence	4	0	Eurostat
BCI_EA	Business Confidence Index	M	Confidence	4	1	OECD
CCI_EA	Consumer Confidence Index	M	Confidence	4	1	OECD
CURR_EACC	Money Stock: Currency	M	Money	2	1	ECB
M1_EACC	Money Stock: M1	M	Money	2	1	ECB
M2_EACC	Money Stock: M2	M	Money	2	1	ECB
WS_EA	Wages and salaries	Q	Labor market	2	2	Eurostat
ESC_EA	Employers' Social Contributions	Q	Labor market	2	2	Eurostat
TEMP_EA	Total Employment (domestic concept)	Q	Labor market	2	2	Eurostat
THOURS_EA	Hours Worked: Total	Q	Labor market	2	2	Eurostat
EMPRT_EA	Quarterly Employment: Wholesale/Retail trade, transport, food	Q	Labor market	2	2	Eurostat
EMPPR_EA	Quarterly Employment: Professional, Scientific, Technical activities	Q	Labor market	2	2	Eurostat
EMPPA_EA	Quarterly Employment: PA, education, health ad social services	Q	Labor market	2	2	Eurostat
GDP_EA	Real Gross Domestic Product	Q	National accounts	2	2	Eurostat
EXPGS_EA	Real Export Goods and services	Q	National accounts	2	2	Eurostat
IMPGS_EA	Real Import Goods and services	Q	National accounts	2	2	Eurostat
GFCE_EA	Real Government Final consumption expenditure	Q	National accounts	2	2	Eurostat
HFCE_EA	Real Households consumption expenditure	Q	National accounts	2	2	Eurostat
CONSD_EA	Real Households consumption expenditure: Durable Goods	Q	National accounts	2	2	Eurostat
GFCF_EA	Real Gross fixed capital formation	Q	National accounts	2	2	Eurostat
GFACON_EA	Real Gross Fixed Capital Formation: Construction	Q	National accounts	2	2	Eurostat
GFAMG_EA	Real Gross Fixed Capital Formation: Machinery and Equipment	Q	National accounts	2	2	Eurostat
MCOILBRENTEU	Brent crude oil spot price	E	Energy commodities	2	1	EIA
PNGASEUUSDM	EU natural gas benchmark price	E	Energy commodities	2	2	IMF
PCOALAUUSDM	Australian coal benchmark price	E	Energy commodities	2	2	IMF
PCOPPUSDM	Global copper price	E	Metals	2	2	IMF
PALUMUSDM	Global aluminum price	E	Metals	2	2	IMF
PNICKUSDM	Global nickel price	E	Metals	2	2	IMF
PIORECRUSDM	Global iron-ore price	E	Metals	2	2	IMF
WB_FOOD	World Bank food price index	E	Food and agriculture	2	1	World Bank
PMAIZMTUSDM	Global corn price	E	Food and agriculture	2	2	IMF
PSOYBUSDM	Global soybean price	E	Food and agriculture	2	2	IMF
PSOILUSDM	Global soybean-oil price	E	Food and agriculture	2	2	IMF
PCOCOUSDM	Global cocoa price	E	Food and agriculture	2	2	IMF
PCOTTINDUSDM	Global cotton price	E	Food and agriculture	2	2	IMF
PRUBBUSDM	Global rubber price	E	Food and agriculture	2	2	IMF
PRAWMINDEXM	Agricultural raw-material price index	E	Food and agriculture	2	2	IMF
GPR	Geopolitical Risk Index	E	Risk and supply chain	4	1	Caldara–Iacoviello
EUEPUINDXM	European Economic Policy Uncertainty Index	E	Risk and supply chain	4	1	Baker–Bloom–Davis
GSCPI	Global Supply Chain Pressure Index	E	Risk and supply chain	4	1	FRBNY

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Table A1: Series in the panel, by information set

Code	Description	Set	Block	TR3	Delay	Source
INDU_006	Industry selling-price expectations	E	Price expectations	4	0	European Commission
RETA_006	Retail price expectations	E	Price expectations	4	0	European Commission
CONS_006	Consumer price expectations	E	Price expectations	4	0	European Commission
FOODPR	Euro-area HICP: processed food incl. alcohol and tobacco	E	HICP subindices	2	2	ECB
FOODUN	Euro-area HICP: unprocessed food	E	HICP subindices	2	2	ECB
IGDSX	Euro-area HICP: non-energy industrial goods	E	HICP subindices	2	2	ECB

Notes: M: monthly; Q: quarterly; E: external monthly. Block, TR3 and Delay as in Table 1. *Source:* EA-MD-QD, release 09.2026, euro-area file (Barigozzi et al., 2026); EIA, IMF and the European policy uncertainty index (Baker et al., 2016) via FRED; World Bank; Caldara and Iacoviello (2022); Benigno et al. (2022); European Commission surveys; ECB.

A.2 Factor Model Diagnostics

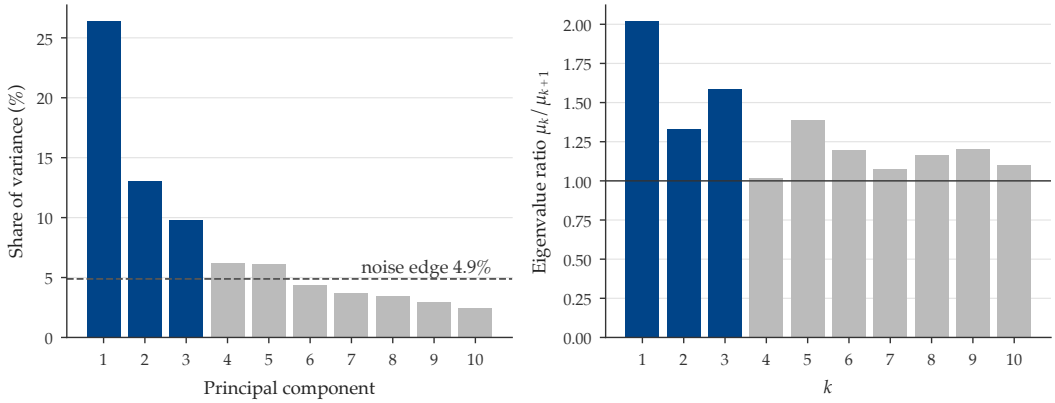


Figure A1: Variance shares and eigenvalue ratios of the leading principal components

Notes: 47 monthly series at the first forecast date. Dark: the three factors used; dashed: noise edge (Marchenko–Pastur).

Table A2: Fit of the factor model by order of the factor VAR

	Log-likelihood, first date	Largest root, first date	Largest root, any date	Dates with the lowest BIC
VAR(1)	−8567.4	0.97	0.98	5
VAR(2)	−8534.1	0.89	0.93	7
VAR(3)	−8517.7	0.90	0.92	0

Notes: EM fits at the first forecast date and each December to 2025 (12 dates). BIC: −2 log-likelihood plus the number of VAR coefficients times log months. The model uses two lags.

Table A3: Lag order of the factor VAR chosen by information criteria

	PCA factors	EM-smoothed factors
AIC	3	5
HQIC	3	3
BIC	3	3

Notes: VAR on the three factors at the first forecast date, up to 6 lags; PCA factors with intercept, EM-smoothed factors without. The model uses two lags.

Table A4: EM fit diagnostics by information set

Set	Fits	Not converged	Median iterations	Maximum iterations	Largest VAR root	Flagged
M	140	0	43	204	0.96	0
MQ	140	0	39	116	0.96	0
ME	140	0	52	126	0.97	0
All	420	0	46	204	0.97	0

Notes: One EM fit per forecast date and set. Flagged: not converged, largest root ≥ 0.98, or numerical failure. *Settings:* stop when the relative change in log-likelihood falls below 10^{-6} or after 3,000 iterations; initial factor covariance 10 times the identity; idiosyncratic variance floor 10^{-8}.

A.3 The Forecasting Ladder in Detail

Table A5: Steps of the forecasting ladder in every information set

Target	Set	FP-PCA / PCA-VAR	FP-DFM / DFM	DFM-AR / DFM	FP-DFM / LASSO	FP-PCA / LASSO
Headline	M	0.972* (0.05)	1.004 (0.86)	0.992 (0.11)	0.893 (0.46)	0.937 (0.45)
Headline	MQ	—	1.003 (0.80)	1.000 (0.31)	0.890 (0.45)	—
Headline	ME	0.977 (0.19)	0.991*** (0.01)	1.000 (0.38)	1.276 (0.37)	1.274*** (0.00)
Energy	M	0.900** (0.03)	1.006 (0.67)	0.949** (0.04)	1.188 (0.39)	1.014 (0.89)
Energy	MQ	—	1.000 (0.58)	0.950* (0.05)	1.253 (0.31)	—
Energy	ME	0.915* (0.10)	1.009 (0.87)	0.958** (0.04)	1.622 (0.15)	1.173** (0.04)
Core	M	0.997 (0.11)	0.953** (0.03)	0.866** (0.04)	0.782* (0.07)	0.895 (0.12)
Core	MQ	—	0.945** (0.02)	0.856** (0.03)	0.820 (0.17)	—
Core	ME	0.996 (0.19)	1.027 (0.95)	0.998 (0.22)	0.798* (0.06)	1.070 (0.38)
Manufacturing	M	0.987** (0.01)	0.998 (0.18)	1.030 (0.83)	1.631*** (0.00)	0.950 (0.62)
Manufacturing	MQ	—	1.000 (0.47)	1.021 (0.97)	1.482*** (0.00)	—
Manufacturing	ME	1.054 (0.17)	1.003 (0.62)	1.038 (0.69)	1.889*** (0.01)	1.079 (0.33)

Notes: MSE of the first model over the second, same set (below 1: the first is better). p-values in parentheses: one-sided Clark–West for nested steps, two-sided Diebold–Mariano against LASSO; * 0.10, ** 0.05, *** 0.01. —: no quarterly series in the PCA models.

Table A6: Steps of the forecasting ladder in M, by period and without 2021–2023

Target	Step	Before 2020	2021–22	2023	2024–25	Full (p)	Outside (p)
Headline	FP-PCA / PCA-VAR	0.2	2.6	0.2	−0.2	2.8* (0.05)	−0.2 (0.41)
Headline	FP-DFM / DFM	0.0	−0.3	−0.1	−0.0	−0.4 (0.86)	0.2 (0.29)
Headline	DFM-AR / DFM	0.3	−0.5	0.4	0.7	0.8 (0.11)	4.6** (0.02)
Headline	FP-DFM / LASSO	−0.5	12.8	0.7	−2.1	10.7 (0.46)	−16.8 (0.24)
Headline	FP-PCA / LASSO	−2.8	11.5	−0.2	−2.2	6.3 (0.45)	−31.4*** (0.00)
Energy	FP-PCA / PCA-VAR	−0.5	6.7	4.1	−0.3	10.0** (0.03)	−2.4 (0.55)
Energy	FP-DFM / DFM	−0.0	0.0	−0.6	−0.0	−0.6 (0.67)	−0.1 (0.68)
Energy	DFM-AR / DFM	−0.1	2.4	2.3	0.5	5.1** (0.04)	1.4 (0.20)
Energy	FP-DFM / LASSO	3.7	−14.3	−8.5	0.3	−18.8 (0.39)	11.7 (0.12)
Energy	FP-PCA / LASSO	−4.3	7.1	−3.1	−1.1	−1.4 (0.89)	−15.7*** (0.01)
Core	FP-PCA / PCA-VAR	0.1	0.1	−0.0	0.1	0.3 (0.11)	0.7 (0.18)
Core	FP-DFM / DFM	1.2	−0.0	3.1	0.5	4.7** (0.03)	5.6*** (0.00)
Core	DFM-AR / DFM	−1.2	0.3	9.3	5.1	13.4** (0.04)	12.6** (0.01)
Core	FP-DFM / LASSO	−0.1	21.2	1.3	−0.6	21.8* (0.07)	−3.0 (0.56)
Core	FP-PCA / LASSO	−0.4	9.3	1.5	0.1	10.5 (0.12)	−1.6 (0.70)
Manufacturing	FP-PCA / PCA-VAR	1.3	0.1	0.1	−0.2	1.3** (0.01)	1.9** (0.02)
Manufacturing	FP-DFM / DFM	−0.0	0.1	0.0	0.1	0.2 (0.18)	0.1 (0.30)
Manufacturing	DFM-AR / DFM	−2.1	−1.5	0.3	0.3	−3.0 (0.83)	−3.2 (0.94)
Manufacturing	FP-DFM / LASSO	−43.4	−25.8	3.0	3.1	−63.1*** (0.00)	−82.4*** (0.00)
Manufacturing	FP-PCA / LASSO	−3.1	5.2	6.6	−3.7	5.0 (0.62)	−14.0 (0.14)

Notes: Set M. Full: percentage MSE reduction of the first model over the second. Period columns: each period's contribution in percentage points, summing to Full. Outside: without 2021–2023. Stars as in Table A5.

A.4 Forecast Robustness

Table A7: Industrial-production robustness: winners and gains over second best

Production	M best	M gain (%)	MQ best	MQ gain (%)
Capital goods	FP-PCA	4.56	AR	3.96
Consumer goods	LASSO	2.15	LASSO	1.94
Durables	AR	15.42	AR	15.93
Non-durables	AR	0.29	AR	0.29
Intermediate goods	PCR	7.98	AR	11.29
Energy	AR	4.48	AR	3.62

Notes: Six industrial-production aggregates, chosen before the run. Gain: percentage MSE reduction of the best model over the second best in the same set. The aggregates overlap.

Table A8: Winning specifications under alternative evaluation samples

Sample	Target	Scope	Best	MSE / AR
All forecasts	Headline	M/MQ/ME	LASSO in ME	0.797
All forecasts	Energy	M/MQ/ME	LASSO in ME	0.951
All forecasts	Core	M/MQ/ME	PCR in ME	0.782
All forecasts	Manufacturing	M/MQ/ME	DFM in MQ	0.305
Without Feb 2022–Dec 2023	Headline	M/MQ	DFM-AR in MQ	0.923
Without Feb 2022–Dec 2023	Energy	M/MQ	DFM-AR in MQ	0.944
Without Feb 2022–Dec 2023	Core	M/MQ	DFM-AR in M	0.898

Notes: Best specification per target on two other samples. All forecasts: all 140 dates, 2020 and 2026 included, M/MQ/ME and shared benchmarks. Without Feb 2022–Dec 2023: also without 2020; inflation targets in M and MQ only.

References

Scott R. Baker, Nicholas Bloom, and Steven J. Davis. Measuring economic policy uncertainty. *Quarterly Journal of Economics*, 131(4):1593–1636, 2016. doi: 10.1093/qje/qjw024.

Matteo Barigozzi, Claudio Lissona, and Lorenzo Tonni. Large datasets for the euro area and its member countries and the dynamic effects of the common monetary policy. *Journal of Applied Econometrics*, 2026. doi: 10.1002/jae.70083. Published online 29 July 2026.

Gianluca Benigno, Julian di Giovanni, Jan J. J. Groen, and Adam I. Noble. The GSCPI: A new barometer of global supply chain pressures. Staff Reports 1017, Federal Reserve Bank of New York, 2022.

Dario Caldara and Matteo Iacoviello. Measuring geopolitical risk. *American Economic Review*, 112(4):1194–1225, 2022. doi: 10.1257/aer.20191823.